$$\prod_{\substack{p \in \mathbb{P} \\ p \leq n}} p \leq 4^n$$

$$\pi(n) = |\mathcal{P}_n| \leq \frac{1{,}3n}{\log n}$$

$$n^{\frac{2n}{\log n}} = e^{\log n \cdot \frac{2n}{\log n}} \leq 4^n$$

$$4^{\log_4 n \cdot \cancel{\log n}} \leq 4^n$$

$$n^{\log n} \leq 4^n = e^{n \log 4}$$

$$e^{\log^2 n}$$

$$\log^2 n \leq n \log 4$$

$$2^{v_2(p+1)} \cdot 2^{v_2(m)} < (p+1)m$$

$$(p+1)(k+1) < 2^l$$

$$(p+1)(k+1) < 1 + p + \dots + p^k$$

$$(p^2-1)(k+1) < p^{k+1} - 1$$

$$k+1 < p^{k-1}$$

$$k \geq 2$$

$$p^{k+1} - p^{k-1} < p^{k+1} - 1$$

$$\binom{pa}{pb} \equiv \binom{a}{b} \pmod{p^3}$$

$$a = \alpha_k p^k + \alpha_{k-1} p^{k-1} + \dots + \alpha_0$$

$$b = \beta_m p^m + \alpha_{m-1} p^{m-1} + \dots + \beta_0$$

$$\binom{\sum_{j=0}^{k} \alpha_j p^{j+1}}{\sum_{j=0}^{m} \beta_j p^{j+1}} = \frac{\left(\sum_{j=0}^{k} \alpha_j p^{j+1}\right)\left(\sum_{j=0}^{k} \alpha_j p^{j+1} - 1\right) \cdots \left(\sum\right.}{}$$

$$k =$$

$$\binom{p^\alpha}{k} = \frac{p^\alpha (p^\alpha - 1) \cdots (p^\alpha - k + 1)}{k(k-1)}$$

$$\frac{p^{\alpha}}{(p^{\alpha-1}+1)(\cancel{p^{\alpha-1}})} \qquad \frac{p^{\alpha}-p^{\alpha-1}-1+1}{1}$$

$p^{\alpha-1}-$

$(p^{\alpha}$